Feature Detection and Extraction Reference

This reading compares different detection and extraction algorithms and summarizes when they should be used. It also lists the default choice of extraction algorithm in MATLAB when using the **`extractFeatures`** function.

Detection and Extraction Algorithms

There are many feature detection algorithms, which can be divided into three main types: corners, blobs, and regions. The proper choice for a detection and extraction algorithm will depend on the specific application and whether it needs to be robust to changes in scale or rotation.

You may refer to the chart below as a reference. This information is also on our documentation page here.

Detection Algorithms

FAST	corner
Min. eigenvalue	corner
Harris	corner
SIFT	blob
SURF	blob
KAZE	blob
BRISK	corner
MSER	region
ORB	corner

Extraction Algorithms

	Scale Invariant	Rotation Invariant
FREAK	✓	✓
SURF	✓	✓
SIFT	✓	✓
BRISK	✓	✓
ORB	X	✓
KAZE	✓	✓
Block	X	X

When you perform feature-based workflows later on, such as image stitching, you can experiment with different algorithms to see which ones produce the best results.

Default Choice of Extraction Algorithm

Recall from the “Extracting Features” video that by default, **`extractFeatures`** uses an extraction algorithm based on the input. The exact choice of extraction algorithm is shown below: